

RAJAVIGNESH J

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PROFESSIONAL SUMMARY

Results-driven Data Analyst and AI Engineer (B.Tech CSE – Data Science & AI, 2026) with 5+ months of internship experience in data analysis, machine learning model development, and full-stack development. Proficient in Python, SQL, Pandas, NumPy, Scikit-learn, TensorFlow, and Power BI. Hands-on experience with ETL pipelines, feature engineering, statistical analysis, NLP, clustering, and deep learning. Delivered data-driven solutions across e-commerce, legal tech, and retail domains. Participated in national-level hackathons including Smart India Hackathon and Amazon ML Challenge.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn

Machine Learning & AI: Scikit-learn, TensorFlow, Keras, Deep Learning, NLP, GANs

Data Visualization: Power BI, IBM Cognos BI, Matplotlib

Databases: MySQL, SQL

Tools & Platforms: Jupyter Notebook, Google Colab, MS Excel

Web Technologies: HTML5, CSS, Flask (basic)

Soft Skills: Analytical Thinking, Problem-Solving, Teamwork, Communication, Attention to Detail

INTERNSHIP EXPERIENCE

Data Analyst Intern — Smarted Innovations, Bangalore

3 Months | 2025

- Cleaned and preprocessed 3 real-world datasets (10,000+ records each) using Python (Pandas, NumPy), reducing data inconsistencies by ~35%.
- Built and evaluated 4 predictive models using Scikit-learn (Logistic Regression, Decision Tree, Random Forest, SVM), achieving up to 88% classification accuracy.
- Designed and delivered 5+ Power BI dashboards to visualize KPIs, enabling faster stakeholder decision-making.
- Developed ETL pipelines to extract, transform, and load data from multiple sources into structured formats for analysis.
- Performed statistical analysis and hypothesis testing to validate business assumptions and derive actionable insights.

Full Stack Development Intern — NoviTech R&D Pvt Ltd, Chennai

1 Month | 2024

- Developed and deployed 3 responsive web application modules using HTML5 and CSS, improving UI consistency.
- Collaborated in an Agile team of 5 to deliver sprint tasks on time, contributing to front-end feature development and bug fixes.

UI/UX Design Intern — Codsoft, Chennai

1 Month | 2024

- Designed 10+ wireframes and interface prototypes following UX best practices, reducing user navigation steps by 20%.
- Conducted user research with 15+ participants and iterated designs based on structured feedback sessions.

PROJECTS

Smart CPC Section Classifier & Civil Case Analyzer (2025)

Tools: Python, NLP, Scikit-learn, TF-IDF, Logistic Regression

- Developed an NLP-based multi-class text classification model to automatically identify relevant CPC sections from legal case descriptions.
- Applied TF-IDF vectorization and Logistic Regression, achieving 91% classification accuracy on a dataset of 2,000+ legal documents.
- Reduced manual legal document review time by an estimated 60% through automated section tagging.

Real or Fake Image Detection (2025)

Tools: Python, TensorFlow, Keras, CNN, GANs, Deep Learning

- Built a GAN-based deep learning model with a CNN discriminator to classify real vs. AI-generated images with 93% accuracy.
- Applied data augmentation on a dataset of 5,000+ images to improve model generalization and reduce overfitting by 18%.
- Deployed the model as a local Flask API for real-time image classification inference.

Customer Segmentation for E-Commerce (2025)

Tools: Python, Pandas, Scikit-learn, K-Means, Power BI

- Applied K-Means clustering to segment 8,000+ customers based on RFM (Recency, Frequency, Monetary) features, identifying 5 distinct customer groups.

- Visualized customer clusters in Power BI dashboards, directly informing targeted marketing strategies that improved campaign relevance by ~25%.

Retail Store Customer Clustering (2025)

Tools: Python, Pandas, NumPy, Unsupervised Learning, Matplotlib

- Performed end-to-end data preprocessing and feature engineering on 12,000+ retail transaction records, handling 15% missing values and outliers.
- Used K-Means and hierarchical clustering to identify 4 customer segments, enabling personalized product recommendations.

T20 World Cup Cricket Data Analytics (2025)

Tools: Python, Pandas, Power BI, JSON, Web Scraping (JavaScript)

- Scraped and ingested T20 World Cup match, batting, bowling, and player data from ESPN Cricinfo using JavaScript web scrapers, producing structured JSON datasets for analysis.
- Built an end-to-end ETL pipeline using Python and Pandas to transform raw JSON files into clean, analysis-ready CSV datasets covering 4 dimensions: match results, player profiles, batting, and bowling summaries.
- Designed an interactive Power BI dashboard to identify the Best XI T20 players, enabling data-driven team selection based on performance metrics such as batting average, strike rate, economy rate, and wickets.

EDUCATION

B.Tech in Computer Science Engineering (Data Science & AI)

Dr. MGR Educational and Research Institute, Chennai | 2022–2026 | CGPA: 7.92 / 10

12th Standard (Higher Secondary Certificate)

V.O.C. Vidiyalayaa MHSS, Chennai | 2022 | 79.3%

CERTIFICATIONS

- IBM Python Fundamentals (2023)
- IBM Java Fundamentals (2023)
- IBM Cloud Application Developer (2024)
- IBM Cognos with Power BI (2024)
- IBM AI & Machine Learning (2025)
- NoviTech Full Stack Development (2024)
- Codsoft UI/UX Design (2024)
- Smarted Innovations – Data Science & Advanced Data Science with Python (2025)

HACKATHONS & COMPETITIONS

- Smart India Hackathon (SIH) – National Level Government Innovation Challenge
- SBI Life HACK-AI-THON – AI/ML Financial Solutions Challenge
- Presidency University Hackathon – Inter-University Tech Competition
- Amazon ML Challenge – Large-Scale Machine Learning Competition